

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

Computer Vision Engineer

I have spent 3.5 years collaborating with cross-functional teams, to ship computer vision solutions on the edge for clients that include British Petroleum and Austin Industries. I own the full-stack: handle large datasets with DVC, build models with PyTorch, track experiments with MLFlow and deploy via Azure. As of June 2025, my systems monitor safety and productivity on construction sites in Texas and on-shore oil rigs in Oman, boosting detection accuracy by 15%.

WORK EXPERIENCE

Esbaar • Muscat, Oman

01/2023 - 08/2023

AI Engineer

- Developed time-series machine learning models for the classification of on-shore oil rig operations to increase worker safety and reduce supervision required during 24/7 operation, achieving a 23% higher F1 score than human experts.
- Built an automated sensor-data cleansing pipeline (3.6 M data points) that cut model roll out from 3 days to less than 24 hours.
- Led strategic MLOps initiatives to continuously enhance experimentation and deployment workflows through bayesian hyperparameter tuning of time-series and vision models.
- Spearheaded the end-to-end development and production rollout of YOLOv5 models, to elevate rig worker and equipment detection precision by 15% in 3 months.
- Refactored a legacy monolithic system into a Kafka-driven microservice architecture, containerising each service with Docker to enhance failure isolation and improve MTBF by 35% within four months
- Designed a dynamic memory management solution to address recurring crashes in a live video streaming system, reducing crash frequency by 67%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

05/2020 - 12/2022

Lead Software Engineer - AI

- Led a 4-engineer team to deliver ViCAS, enhancing model accuracy and reducing false positives by 33% through effective framing and edge-case resolution within 6 months.
- Implemented real-time object tracking with YOLO-DeepSORT, maintaining consistent identities across frames and enabling reliable downstream analytics.
- Deployed quantised OpenVINO vision models over Microsoft Azure to rugged Intel VPU edge devices for on-site inference in harsh industrial environments.
- Accelerated vision model inference with mixed-precision quantisation and problem-reframing, reducing latency from 4.5 FPS to 15 FPS, enabling real-time inference and reporting of OSHA violations.
- Repurposed legacy object detection models for automated labelling pipeline workflows, reducing manual annotation effort by 80% and accelerating dataset curation.
- Engineered synthetic training data pipelines using OpenCV and NumPy, reducing edge-case model development time from 6 days to 2 days for high-risk production environments.

EDUCATION

Master of Science in Robotics - 2:1

University of Bristol • Bristol, England

09/2023 - 09/2024

Engineered a UGV system (Clearpath Husky) with multi-sensor fusion (Velodyne LiDAR, Intel RealSense RGBD camera, Symetrical Verifinder).

Deployed real-time LiDAR SLAM (ROS2 Nav2 + Velodyne) on Jetson AGX Xavier, containerised with Docker for field-deployable, point-cloud mapping.

Achieved precise radionuclide localisation using advanced odometry techniques like Scan-to-Map ICP, by significantly reducing drift in real-time LiDAR SLAM operations.

Implemented a mono camera calibration and point-cloud 3-D reconstruction pipeline in C++/OpenCV for the M.Sc. Robotics thesis, delivering metric-accurate environment mapping for downstream perception tasks.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

07/2016 - 05/2020

CERTIFICATIONS

Deep Learning Specialisation

deeplearning.ai

07/2019

AWARDS & SCHOLARSHIPS

Best Speaker award

ABC Toastmasters club

SKILLS

- **Perception & Robotics:** LiDAR SLAM, Nav2, OpenCV, OpenVINO, ROS2, Vision Transformer (ViT), YOLOv5/v8
- **Backend & Cloud / Edge:** Apache Kafka, AWS, FastAPI, Intel VPU, Microsoft Azure, NVIDIA Jetson AGX, PostgreSQL
- **DevOps & MLOps:** Docker, DVC, Git, Linux, MLFlow, Optuna